



PRODUCT NAME

Industrial-strength Powder Detergent Formula

Description

laundry powder detergent

Technical Specifications

pH Level:	10
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Carbonate	Sodium Carbonate	24.51%	Builder and pH adjuster
2	Sodium Sulfate	Sodium Sulfate	34.31%	Filler
3	Sodium Tripolyphosphate	Sodium Tripolyphosphate	11.76%	Water softener and builder
4	Linear Alkylbenzene Sulfonate (LAS) 90%	Sodium C10-13 Alkyl Benzene Sulfonate	14.71%	Primary surfactant
5	Sodium Silicate	Sodium Silicate	6.86%	Corrosion inhibitor and alkalinity source
6	Sodium Percarbonate	Sodium Carbonate Peroxide	4.9%	Oxygen bleach and stain remover
7	Enzyme Blend	Amylase, Protease	1.96%	Stain removal
8	Optical Brightener	Disodium Distyrylbiphenyl Disulfonate	0.49%	Whitening
9	Anti-caking Agent	Silica	0.5%	Flow enhancement
Total			100%	

Manufacturing Process

Phase 1: Preparation

- Weigh all ingredients according to the formulation table
- Ensure all equipment is clean and sanitized
- Set up mixing equipment at appropriate temperature

Phase 2: Main Processing

- Add water phase ingredients to mixing vessel
- Begin stirring at medium speed
- Gradually add active ingredients while maintaining constant mixing
- Monitor temperature and pH throughout the process

Phase 3: Final Processing

- Add preservatives and adjust pH if necessary
- Continue mixing until homogeneous
- Perform quality control checks
- Package in appropriate containers

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions